

SEAN KHOMPHENGCHAN

CURRENT ADDRESS
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SOCIALS
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OBJECTIVE

To obtain an engineering Full-Time position in a Guidance, Navigation, & Controls or Software related position applicable to the space sector.

EDUCATION

Purdue University, West Lafayette, IN December 2023
Bachelor of Science in Aeronautical & Astronautical Engineering
Specializations: Dynamics and Control
GPA: 3.16 / 4.00
Relevant Coursework: Spacecraft Attitudes, Multi Agent Autonomy, Applied Controls, Satellite Navigation and Positioning
Technical Skills: MATLAB, Simulink, Python, C, C++, Stability Analysis, Kalman Filtering, Convex Optimization, Git

TECHNICAL & RELEVANT EXPERIENCE

Simulation and Modeling Intern June 2022 – August 2023
Raytheon, Arlington VA

- Developed algorithms to predict and simulate platform performance and to conduct threat assessments.
- Briefed executives all the way to the president of the company on work accomplished and impact of.
- Directly contributed to a F-35 engine study that led to the winning of a \$5.2-billion contract for an engine upgrade.
- Utilized Object Oriented Programming to build campaign level simulation software.
- Used Monte-Carlo Simulations to analyze effectiveness of different methods and solutions.

Undergraduate Teaching Assistant August 2021 – May 2023
Purdue University, West Lafayette IN

- Provided guidance to over 100 students on MATLAB, Python, Excel to refine course concepts.
- Piloted a successful computer lab for students to access help for bridging programming and engineering.
- Assisted in design of coursework and curriculum for First Year Engineering.
- Graded for Control Systems Design in addition to supporting the First Year Engineering Department.

Undergraduate Research Assistant at Ma Quantum Lab January 2021 – June 2021
Purdue University, West Lafayette IN

- Created and presented a poster using Latex at the Purdue Undergraduate Research Conference.
- Researched Microwave electronics for superconducting quantum circuits for use on quantum simulation testing.
- Realized noise reduction of 90% on a log curve when trying to analyze signal received from system.
- Collaborated with Principal Investigator and lab group to exchange idea for finding better methods for end-to-end testing of system.

PROJECTS

Convex Six-DOF Spacecraft Model January 2022 – May 2023

- Incorporated Kalman Filters into control system design of active controlled model rocket
- Create and model motion of spacecraft using sensor data with MATLAB and Simulink
- Convexified trajectory and attitude of spacecraft using CVX
- Successfully simulated model rocket reorienting and self-landing

Filtering Analysis for GNSS Signal Correction August 2022 – December 2022

- Semester long research project in conjunction with Satellite Navigation Course.
- Utilizes a variety of different verified Filtering models for signal analysis of Satellite Navigation Systems.
- From independently sourced GNSS data, was able to get an accuracy of 5 meters and low mean square root error.
- Built mainly utilizing MATLAB and Python.
